A Viscous-Drag Governor: Automatic Steam-Engine Speed Regulation by Fluid-Shear Sensing

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***Abstract*—The ungoverned steam engine of the late eighteenth century loses or gains speed in almost direct proportion to the mechanical load on its line shaft, and can therefore overspeed to destruction or stall. Every historical solution—from Watt's flyball to the Porter, Proell and Hartnell governors—senses speed through the centrifugal force of rotating masses, a force proportional to ω^2. This paper invents a different sensing principle: a *viscous-drag governor* (VDG) in which a rotor carried by the engine shaft shears a thin oil film that drags a concentric, spring-restrained sensing drum. The shear torque is linear in the slip speed, so the drum deflection is linear in ω rather than quadratic and the loop gain is independent of operating speed. The mechanism adds only an oil film, a torsion spring and a short lever linkage—five plain turning pairs, all of them makeable in a 1780s workshop. A first-principles nonlinear model is derived and linearised about the normalised operating point ($\omega_0 = 1$, $T_{L0} = 1$), giving $G_p(s)$, $G_c(s)$ and a fourth-order characteristic polynomial. Routh–Hurwitz analysis and the Bode margins (PM $= 66.3^\circ$, GM $= 12.0$ dB) confirm stability with a dominant pair at $\omega_n = 3.62$ rad/s, $\zeta = 0.671$. Simulated $\pm 30\%$ load steps cut the steady-state speed error from 30% to 1.76%, with a peak deviation of 1.85% and a settling time of 1.58 s. Adding a viscous reset dashpot drives the steady-state error to zero while retaining a 57.1° phase margin.**

***Index Terms*—Governor, steam engine, speed regulation, viscous shear, fluid-shear sensing, linearisation, Routh–Hurwitz, transient response, disturbance rejection.**

I. Introduction

A. Industrial Context and Problem Statement

By the 1780s a single steam engine drove an entire cotton mill through a common line shaft and belt system. Individual machines were engaged and disengaged at will, so the mechanical load changed abruptly and unpredictably. Because the throttle was set by hand, the engine could not hold its speed: a heavy load slowed it to a stall, and a machine disengaged overspeeded it, risking flywheel burst and the destruction of the shafting [3]. The problem is one of *disturbance rejection*: hold the shaft speed at its nominal value under large, unmeasured steps in load torque, using only late-eighteenth-century mechanisms.

Source code, figures and reproduction instructions: https://github.com/ZakoAtIt/me319-viscous-drag-governor

B. Design Objectives

This work invents a new, fully automatic speed regulator and analyses it as a linear feedback system. The objectives are: (i) a sensing principle physically distinct from the centrifugal sensing of every historical governor; (ii) a mechanism that could actually have been built in the 1780s; (iii) a rigorous first-principles model, linearised about the nominal point; (iv) closed-loop stability and transient-response analysis with standard control tools; and (v) open-loop versus closed-loop simulation of $\pm 30\%$ step loads at $\omega_0 = 1$, $T_{L0} = 1$.

C. Assumptions

Physical. (1) Small-perturbation linear dynamics about the nominal operating point are valid, as permitted by the brief. (2) Indicated torque is proportional to throttle opening at constant cutoff. (3) The engine and flywheel inertia is one lumped inertia J, and the whole resisting torque is the mill load $T_L(\omega)$ of Eq. (2); its slope at the nominal point *is* the effective viscous damping b, which therefore does not appear separately in Eq. (3). (4) The throttle valve is a first-order lag of time constant τ_v. (5) The oil film is laminar and isothermal—the gap Reynolds and Taylor numbers are far below the instability thresholds, and the film dissipates only $c_s\omega_0^2 \approx 80$ W, under 0.5% of engine power—so the shear stress obeys Newton's law of viscosity [6] and is linear in the slip speed. (6) Linkages are rigid, massless and backlash-free in the nominal model; backlash and friction are treated explicitly as failure modes in Sec. VI. (9) The nominal throttle setting $x_{v0} = 1$ is half open, giving ample headroom to reach $x_v = 1.28$ without striking the wide-open stop. **Historical.** (7) No governor is assumed to exist. (8) Only 1780s materials and processes are permitted—cast and wrought iron, brass, wood, leather and vegetable oil; electrical, magnetic and elastomeric components are excluded by construction, which is why the conventional eddy-current "drag-cup" sensor is not used.

II. Conceptual Design

A. The Sensing Principle

The proposed mechanism replaces the centrifugal flyweights with a *viscous shear element*: a shallow *shear rotor* keyed to the engine shaft runs inside a concentric, sealed oil chamber (Fig. 1) whose wall is the *sensing drum*. The drum is not keyed to the shaft; it is free to rotate through a small angle about the same axis, restrained by a calibrated torsion spring anchored to the frame. (The geometry is unrelated to the eddy-current "drag-cup" tachometer of later centuries, where the coupling is electromagnetic rather than viscous.)

At constant speed the rotor drags the oil, the oil drags the drum, and the drum comes to rest at the angle at which the transmitted shear torque is exactly balanced by the spring. Because the film flow is laminar, the shear torque is linear in the slip speed,

$$T_{\text{shear}} = \frac{\mu A r^2}{h}\left(\omega - \dot{\theta}\right) = c_s\left(\omega - \dot{\theta}\right), \tag{1}$$

where μ is the dynamic viscosity, A the wetted area, r the mean radius and h the film thickness. The steady deflection $\theta_{ss} = (c_s/k_s)\,\omega$ is therefore *linear in speed*, unlike the ω^2-dependent lift of a flyball governor. With Table I's parameters it is 72° at rated speed, and a 30% load step moves the drum 1.3°—easily resolvable. The drum carries a lever linked to the steam throttle; because the drum rotates further as the engine speeds up, the linkage is arranged to *close* the throttle as the deflection grows, giving negative feedback.

B. Why This Is Not a Known Governor

Fundamental novelty. Watt's flyball and its Porter, Proell, Hartnell, Wilson–Hartnell and Pickering descendants [2] all sense speed through centrifugal force, proportional to ω^2, and differ only in the geometry of the mass–link–sleeve assembly. The VDG removes the rotating masses entirely and senses speed through the viscous shear stress of an oil film—a different physics, a different sensor law (linear, not quadratic) and a different set of failure modes. Three consequences follow that are difficult to obtain from a flyball governor.

1) *Weakly speed-dependent loop gain.* A flyball's sensing force scales as ω^2, so its incremental gain at a given speed scales as ω: it changes by a factor of ten between 10% and 100% of rated speed. The VDG's sensing torque scales as ω^1, so its incremental gain is independent of speed and the loop gain $K_t K_l c_s/k_s$ derived in Sec. III stays constant across the working range. (The engine's own damping b is also speed-dependent, so this advantage is one of degree rather than absolute.)
2) *Fewer moving parts and fewer wear surfaces.* The sensor needs one extra bearing—the drum, free on the engine shaft—and a three-bar lever linkage, five plain turning pairs in all, every one of them an oiled pin or a plain bearing. A flyball assembly needs flyweights, a sleeve, links and their pivots, the sliding sleeve in particular being the part that dominates the wear of centrifugal governors.
3) *No small-arc restriction.* The flyball governor is inherently nonlinear and behaves well only while its flyweights move over a small arc. The VDG's sensing law, Eq. (1), is linear over its whole travel, which is what makes the constant-gain claim above hold. The remaining nonlinearities—the speed-dependent drag of the driven machinery and the valve flow–lift law—are mild and are linearised in Sec. III.

Similarities to existing governors, stated candidly. The loop architecture is deliberately *the same* as Watt's: a speed sensor drives a lever and link that positions the steam throttle, with the sense arranged so that rising speed closes the valve. In control terms the sensor is the feedback element $H(s) = c_s/(J_s s^2 + c_s s + k_s)$, and $G_c(s)$ is the equivalent controller obtained by grouping H with the valve, a grouping legitimate only because the setpoint is constant. The comparator, the actuator and the feedback structure are not new, and no claim is made that they are: what is new is the *sensing element*.

Related mechanisms from the literature are the *eddy-current drag-cup tachometer*, abandoned here because it is electromagnetic and so excluded by the brief; the *viscous damper* and *fluid coupling*, whose torque is transient and decays to zero at constant speed whereas this film develops a *steady* shear stress proportional to speed; and the hydraulic-relay and pneumatic servo governors of later centuries, which use fluid *power amplification* but retain a centrifugal or clock-driven *sensor*. Mechanical integral ("reset") control is likewise not new—it is a nineteenth-century development [2]—and appears here only as an optional variant (Sec. III). The torsion spring is essential: without it the drum would simply be entrained by the oil, since the film's shear stress can only be converted into a position by balancing it against a known stiffness.

The four elements of Fig. 1 are (i) the shear rotor, oil chamber and sensing drum, converting speed into the angle θ; (ii) the torsion spring, whose stiffness sets the speed–angle gain c_s/k_s and anchors the drum to the frame; (iii) the drum lever, link and throttle lever, which convert θ into a throttle motion of gain $K_l = \mathrm{d}x_v/\mathrm{d}\theta$ and reverse its sense—the link attaches on one side of the pivot and the valve stem hangs from the other, so rising speed lowers the stem and closes the valve; and (iv) the light, stiff throttle valve, a first-order lag τ_v. The linkage is a four-bar chain grounded at the drum bearing and the throttle-lever pivot, plus the valve-stem joint: five turning pairs in all, all visible in Fig. 1.

C. Closed-Loop Block Diagram

Fig. 2 shows the loop as it is actually wired. The viscous sensor lies in the feedback path and produces the drum angle θ; the *difference* $e = \theta_{\text{ref}} - \theta$ between it and the setpoint (the calibrated stop on the valve lever) positions the throttle through $G_c(s) = K_l c_s/(J_s s^2 + c_s s + k_s)$ and the valve lag. The steam torque drives the shaft inertia against viscous and external loading, giving $G_p(s) = K_t/[(Js + b)(\tau_v s + 1)]$; the load torque T_L enters directly at the shaft and is the disturbance to be rejected.

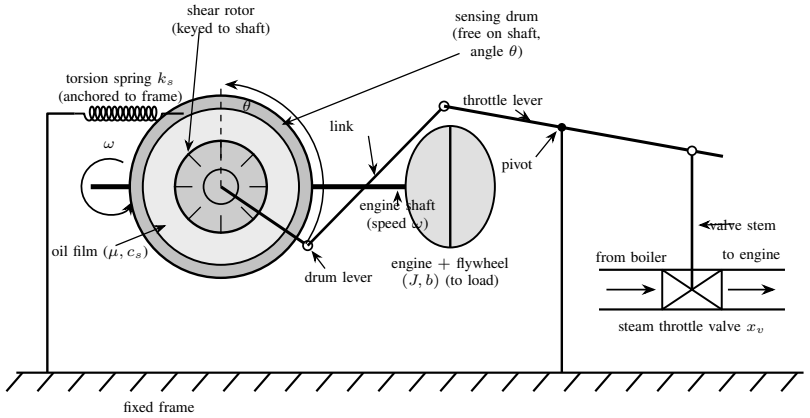


Fig. 1. Engineering schematic of the viscous-drag governor. A shear rotor keyed to the engine shaft shears the oil film in the sealed chamber; the shear torque is balanced by a torsion spring anchored to the frame, so the sensing drum (free on the shaft) settles at a deflection θ proportional to engine speed. The drum lever and link transmit θ to the throttle lever, and the link and the valve stem lie on *opposite* sides of the pivot, so the stem descends and closes the steam valve as speed rises ($K_l = \mathrm{d}x_v/\mathrm{d}\theta$). Counted from the drawing, the linkage has five plain turning pairs: the drum bearing, the two link pins, the throttle-lever pivot and the valve-stem joint.

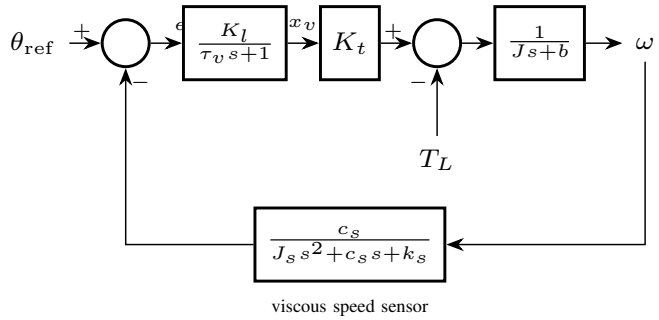


Fig. 2. Closed-loop block diagram. The viscous sensor sits in the *feedback* path, because an oil-film drum measures shaft speed ω and cannot sense the error directly; it produces the drum angle θ, compared with the setpoint θ_{ref} set by the calibrated stop on the valve lever. The load torque T_L enters *after* the throttle gain K_t, at the shaft, as Eq. (14) requires. Grouping the sensor with the valve gives the equivalent controller $G_c(s)$ of Eq. (11), with $G_p(s) = K_t/[(Js+b)(\tau_v s+1)]$.

III. Mathematical Modeling

A. Nonlinear Governing Equations

Let ω be the shaft speed. The driven machinery of a mill is a mixture of constant-torque machines (looms, hammers) and speed-dependent machines (fans, pumps, and the viscous drag of belts and journals), so the load torque is written as the nonlinear function

$$T_L(\omega) = T_a + c_1\omega + c_2\omega^2, \tag{2}$$

in which the constant T_a represents looms and hammers, the linear term $c_1\omega$ journal and belt friction, and the quadratic term $c_2\omega^2$ fans and pumps; $T_e(t)$ is the switched torque of a machine engaged or disengaged—precisely the disturbance the governor must reject. With a constant-cutoff steam admission the indicated torque is proportional to throttle opening, and the shaft equation is

$$J\dot{\omega} = K_t\, x_v - T_L(\omega). \tag{3}$$

The sensing drum obeys angular momentum balance with the film torque of Eq. (1) and the spring torque,

$$J_s\ddot{\theta} + c_s\dot{\theta} + k_s\theta = c_s\,\omega, \tag{4}$$

and the throttle valve is a first-order lag driven by the linkage,

$$\tau_v\dot{x}_v + x_v = K_l\big(\theta_{\mathrm{ref}} - \theta\big). \tag{5}$$

Equations (3)–(5) are the nonlinear model. The nonlinearity resides in $T_L(\omega)$ (quadratic drag of the driven machinery) and would also appear in a nonlinear valve flow–lift law.

B. Linearisation

Choose the nominal operating point $\omega_0 = 1$, $T_{L0} = T_L(\omega_0) = 1$, $x_{v0} = 1$ and $\theta_0 = c_s\omega_0/k_s$. Writing $\omega = \omega_0 + \delta\omega$ and similarly for the other variables, and expanding Eq. (2) to first order, the load torque perturbation is $\delta T_L = b\,\delta\omega + \delta T_{L,e}$, where

$$b \;\triangleq\; \left.\frac{\mathrm{d}T_L}{\mathrm{d}\omega}\right|_{\omega_0} = c_1 + 2c_2\omega_0 \tag{6}$$

is the *effective viscous damping* and $\delta T_{L,e}$ is the external step disturbance. For the design used here $T_a = 0.25$, $c_1 = 0.5$ and $c_2 = 0.25$, so that $T_L(\omega_0) = 1$ and $b = c_1 + 2c_2\omega_0 = 1$: this is the physical origin of b, which lumps journal friction together with the velocity-dependent drag of the mill. Equation (3) at the nominal point then requires $K_t x_{v0} = T_{L0} = 1$, so with $x_{v0} = 1$ the throttle gain is $K_t = 1$—the damping torque is *inside* T_L and must not be counted a second time in the steady-state balance. The linearised system is therefore

$$J\,\delta\dot{\omega} = K_t\,\delta x_v - b\,\delta\omega - \delta T_{L,e}, \tag{7}$$

$$J_s\,\delta\ddot{\theta} + c_s\,\delta\dot{\theta} + k_s\,\delta\theta = c_s\,\delta\omega, \tag{8}$$

$$\tau_v\,\delta\dot{x}_v + \delta x_v = -K_l\,\delta\theta. \tag{9}$$

Note that Eq. (8) is already linear: the VDG needs no linearisation of its own sensing element.

C. Plant and Controller Transfer Functions

Taking Laplace transforms of Eqs. (7)–(9), the plant is

$$G_p(s) = \frac{\delta\omega}{\delta x_v} = \frac{K_t}{(Js+b)(\tau_v s+1)} = \frac{1}{(10s+1)(0.1s+1)}, \tag{10}$$

and the controller, taken from the speed error $e = \omega_{\mathrm{ref}} - \omega$ to the commanded throttle, is

$$\begin{aligned} G_c(s) = \frac{\delta x_v}{e} &= \frac{16}{0.02\,s^2 + 0.2\,s + 1}, \\ &= \frac{K_l\,(c_s/k_s)}{(J_s/k_s)\,s^2 + (c_s/k_s)\,s + 1}. \end{aligned} \tag{11}$$

With the parameters of Table I the DC loop gain is $G_0 = G_c(0)G_p(0) = K_t K_l c_s/(k_s b) = 16$.

D. Closed-Loop Transfer Functions

The loop transfer function is

$$\begin{aligned} L(s) &= G_c(s)G_p(s) \\ &= \frac{16}{0.02s^4 + 0.402s^3 + 3.04s^2 + 10.3s + 1}, \end{aligned} \tag{12}$$

so the closed-loop transfer function from the speed setpoint to the speed is

$$\begin{aligned} G_{CL}(s) = \frac{\omega(s)}{\omega_{\mathrm{ref}}(s)} &= \frac{L(s)}{1+L(s)} \\ &= \frac{16}{0.02s^4 + 0.402s^3 + 3.04s^2 + 10.3s + 17}. \end{aligned} \tag{13}$$

Because the load torque is applied directly at the shaft, the disturbance transfer function is different and is the quantity that governs the *regulation* problem:

$$\begin{aligned} \frac{\omega(s)}{T_L(s)} &= \frac{-1/(Js+b)}{1 + G_c G_p} \\ &= \frac{-(0.1s+1)\,(0.02s^2 + 0.2s + 1)}{0.02s^4 + 0.402s^3 + 3.04s^2 + 10.3s + 17}. \end{aligned} \tag{14}$$

The DC gain of Eq. (14) is the speed droop of a type-0 loop [5],

$$\text{droop} = \left|\frac{\omega(\infty)}{T_L(\infty)}\right| = \frac{1}{b\,(1+G_0)} = \frac{1}{17} = 5.88\%, \tag{15}$$

i.e. a full (100%) change of load torque produces a 5.88% permanent speed change. For the $\pm 30\%$ steps specified in the brief this is only $\pm 1.76\%$.

E. Reset (Integral) Variant

The residual droop of Eq. (15) can be removed by a standard mechanical *reset*: a viscous dashpot is inserted between the linkage and the spring anchor so that a persistent deflection slowly re-trims the spring seat until the error vanishes. This multiplies Eq. (11) by the factor $(T_i s+1)/(T_i s)$ with $T_i = 4$ s, which contains an integrator and therefore forces zero steady-state error. The resulting fifth-order characteristic polynomial, from $1 + G_c G_p = 0$, is $0.08s^5 + 1.608s^4 + 12.16s^3 + 41.2s^2 + 68s + 16 = 0$.

IV. Control and Stability Analysis

A. Routh–Hurwitz Analysis

Closed-loop stability is the classical problem first posed by Maxwell [1]. Substituting Eqs. (11) and (10) into $1 + G_c G_p = 0$ gives the closed-loop characteristic polynomial

$$0.02s^4 + 0.402s^3 + 3.04s^2 + 10.3s + 17 = 0. \tag{16}$$

The Routh first column is $(0.02,\ 0.402,\ 2.528,\ 7.596,\ 17)$, all strictly positive, so by the Routh–Hurwitz criterion [4] the closed loop is asymptotically stable; the binding s^1 element, $c_1 = (b_1 a_1 - a_3 a_0)/b_1 = 7.60 > 0$, confirms it. The same array gives the gain at which stability is lost: $G_0^{\mathrm{crit}} = 63.8$, or 3.99 times the design value of 16, in agreement with the 12.0 dB gain margin found below.

B. Closed-Loop Poles

Solving Eq. (16) gives

$$s_{1,2} = -2.432 \pm j2.685, \quad s_{3,4} = -7.618 \pm j2.595 \ \ [\mathrm{rad/s}]. \tag{17}$$

The pair $s_{1,2}$ lies closest to the imaginary axis and therefore dominates the response, with $\omega_n = 3.62$ rad/s and $\zeta = 0.671$. This damping ratio corresponds to an equivalent second-order overshoot of 5.8%; the observed 4.6% (Sec. V) is slightly smaller because the distant pair $s_{3,4}$ ($\zeta \approx 0.95$) adds a small in-phase component. The dominant real part (-2.43) sets the response speed: a loop time constant of 0.41 s against the plant's $J/b = 10$ s, i.e. feedback makes the governed engine about $20\times$ faster while cutting its load sensitivity by a factor of 17.

C. Frequency-Response Margins

The loop transfer function (12) has its gain crossover at $\omega_c = 1.575$ rad/s, giving

$$\mathrm{PM} = 66.3^\circ, \qquad \mathrm{GM} = 12.0\ \mathrm{dB} \text{ at } 5.06\ \mathrm{rad/s}. \tag{18}$$

The margin remains sufficient because the crossover lies well below both the sensing-drum natural frequency ($\omega_{n,s} = \sqrt{k_s/J_s} = 7.07$ rad/s) and the valve corner ($1/\tau_v = 10$ rad/s), so little phase lag enters inside the loop bandwidth. This is the central design rule of the VDG: *the drum must be stiff and light relative to the loop bandwidth.*

D. The Reset Variant

For the reset design of Sec. III the closed-loop poles are

$$-7.497 \pm j2.537, \quad -2.413 \pm j2.374, \quad -0.279 \ \ [\mathrm{rad/s}], \tag{19}$$

with PM $= 57.1^\circ$ and GM $= 11.6$ dB; the Routh first column $(0.08, 1.608, 10.11, 30.51, 61.90, 16)$ is entirely positive, so the reset loop is also stable. The real pole at -0.279 is dominant and reflects the reset time constant T_i, the price paid for zero droop: integral action removes the offset but costs 9° of phase margin and slows the final settling.

V. Simulation and Results

A. Method and Parameters

The model was implemented in MATLAB R2025b (Octave-compatible scripts; repository link in the footnote) and integrated with a classical fourth-order Runge–Kutta scheme, fixed step 2 ms over 80 s, so no toolbox function is required. Step disturbances from $T_L = 1$ to 1.3 and to 0.7 were applied; the open-loop case is the same plant with the throttle held at $x_{v0} = 1$.

Table I lists the parameters with their dimensional anchors. A nominal speed of 60 rpm and power of 20 kW give 3183 N·m of torque, so $J = 10$ represents a 2 t flywheel of 1.6 m radius of gyration—typical of a mill engine—and J_s a 1.3 kg drum shell of 0.40 m radius, consistent with Fig. 1. The linkage gain $K_l = 80$ is realised by a 0.40 m drum radius feeding a 20:1 compound lever.

B. Open-Loop Versus Closed-Loop Response

Fig. 3 compares the two cases. Without a governor a $+30\%$ load step drives the speed from 1 to 0.700, a 30% steady-state error reached with the plant's slow 10 s time constant; the -30% step is exactly symmetric, with the speed rising to 1.300. This near-proportional load–speed relationship is the historical failure mode of the brief and follows directly from the normalisation $b\,\omega_0 = T_{L0}$.

With the VDG closed the same $+30\%$ step gives a peak deviation of only 1.85% ($\omega = 0.9815$) and a steady-state error of 1.76% ($\omega_\infty = 0.9824$), reached after 1.58 s (2% of the final change); the -30% step is symmetric ($\omega_\infty = 1.0176$). The response is monotonic to within the band, with a 4.6% overshoot relative to its final value, consistent with $\zeta = 0.671$, and the dominant-pair angle of 48° gives $\zeta = \cos 48^\circ = 0.67$ as computed analytically. The throttle moves 0.282 normalised units (28% of the nominal opening, within the half-open starting position of Sec. I), driven by the 1.3° drum rotation predicted by $\delta\theta = (c_s/k_s)\,\delta\omega$.

C. Effect of the Reset

Table II contrasts the proportional design with the reset (PI) design and collects all metrics. The reset drives the steady-state error to zero and the peak deviation slightly *down*, from 1.85% to 1.77%: adding $1/(T_i s)$ raises the loop gain at every frequency below crossover, so the reset response is bounded in magnitude by the proportional one. The price is a longer tail: the reset enters the $\pm 0.5\%$ band at 5.5 s, against 1.58 s for the proportional design to enter its $\pm 2\%$ band. Which design is preferred depends on the duty: the proportional version is the simpler and slightly better damped mechanism and is the one recommended for the mill drive, where a 1.76% speed change under a 30% load step is already an order of magnitude better than anything a 1780s engine could otherwise achieve; the reset version is recommended where several engines must run in parallel and be synchronised, since integral action removes the standing droop that would otherwise accumulate into a phase error.

TABLE I
Model parameters. Normalised values are those of Eqs. (2)–(17); physical values follow from the scales $\tau_0 = 1$ s, $\omega_0 = 6.283$ rad/s (60 rpm), $T_0 = 3183$ N·m (20 kW) and $\theta_0 = \omega_0\tau_0$.

Symbol	Norm.	Physical	Basis
J	10	5.1×10^3 kg·m^2	flywheel + engine; spin-down time $J/b = 10$ s
b	1	507 N·m·s/rad	slope of T_L at ω_0; $b\omega_0 = T_{L0}$
K_t	1	—	nominal balance $K_t x_{v0} = T_{L0} = 1$
τ_v	0.1 s	0.1 s	valve lag; ≈ 1.6 Hz bandwidth of the stem + lever
J_s	4×10^{-4}	0.20 kg·m^2	sensor time constant $J_s/k_s = 0.02$ s^2
c_s	4×10^{-3}	2.0 N·m·s/rad	film transmits 12.7 N·m (80 W) at ω_0
k_s	0.02	10.1 N·m/rad	$\theta_{ss} = 72^\circ$ full scale, 1.3° per load step
K_l	80	—	linkage gain; sets $G_0 = 16$
T_i	4 s	4 s	reset: shortest tail that still keeps PM $> 55^\circ$

J, b, K_t, τ_v follow from the stated engine and the nominal torque balance; the sensor and linkage entries were *fitted* to meet the design targets, their absolute scale then being set by the drum size and the film-dissipation limit (appendix).

TABLE II
Transient-response metrics for $\pm 30\%$ step load disturbances (normalised variables, $\omega_0 = 1$, $T_{L0} = 1$)

Case	ω_∞	err. [%]	peak [%]	t_s [s]	$x_{v\infty}$
$+30\%$ *load step* ($T_L : 1 \to 1.3$)					
Open loop	0.700	30.0	30.0	38.96	1.000
VDG (P)	0.982	1.76	1.85	1.58	1.282
VDG (P+reset)	1.000	0.00	1.77	5.5	1.300
-30% *load step* ($T_L : 1 \to 0.7$)					
Open loop	1.300	30.0	30.0	38.96	1.000
VDG (P)	1.018	1.76	1.85	1.58	0.718
VDG (P+reset)	1.000	0.00	1.77	5.5	0.700

t_s: time to enter the $\pm 2\%$ band about ω_∞; for the reset rows, whose final error is zero, the $\pm 0.5\%$ band. For the open loop "peak" is the final error (monotone response), for the governor the transient excursion. The reset peak is below the P peak because integral action only raises the loop gain.

VI. Conclusions

A. Summary

A new speed governor has been invented in which the centrifugal flyweights of every historical governor are replaced by a viscous-shear sensing element: a rotor shears an oil film that deflects a spring-restrained sensing drum, giving a speed signal *linear* in ω rather than quadratic. A first-principles nonlinear model was derived and linearised about the normalised operating point, yielding $G_p(s)$, $G_c(s)$, $G_{CL}(s)$ and a fourth-order characteristic polynomial. Routh–Hurwitz analysis proves asymptotic stability (first column $0.02, 0.402, 2.528, 7.596, 17$) with a phase margin of 66.3°, a gain margin of 12.0 dB and a dominant pair at $\omega_n = 3.62$ rad/s, $\zeta = 0.671$. Under the prescribed $\pm 30\%$ load steps the governor cuts the speed error from 30% to 1.76%, with a 1.85% peak deviation and a 1.58 s settling time, while a viscous reset drives the error to zero.

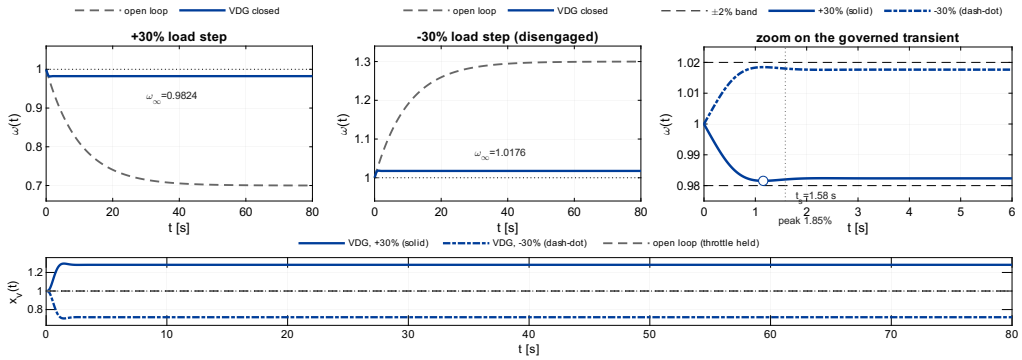


Fig. 3. Open-loop versus closed-loop response to the prescribed load steps. Top row: engine speed $\omega(t)$ for a $+30\%$ step, for a -30% step (machine disengaged), and a zoom on the governed transient that makes the 1.85% peak and the 1.58 s settling time visible. Bottom: throttle position $x_v(t)$. The ungoverned engine changes speed by 30% and takes ≈ 39 s to settle. Colour is fixed across every panel: grey dashed = open loop, blue = governed; the sign of the step is given by the title or the line style.

B. Advantages

1) The sensing law is linear, so the loop gain is independent of working speed and the governor behaves almost identically at low and high speed, which a flyball governor can only approximate.
2) Fewer moving parts: the sensor adds one plain bearing and a three-bar linkage—five turning pairs (Fig. 1)—whereas a flyball assembly needs flyweights, a sliding sleeve, links and their pivots, the rubbing sleeve being the part that dominates its wear.
3) The oil film is self-damping, so the sensor adds useful loop damping rather than the lightly damped resonance of a spring–flyweight system.
4) Every component—cast-iron drum, brass chamber, steel torsion spring, leather seals, vegetable oil—was available to an eighteenth-century instrument maker.

C. Limitations and Failure Modes

1) **Viscosity drift—the dominant limitation.** The gain $c_s = \mu A r^2/h$ is proportional to μ, and a rise in oil temperature from 15°C to 50°C lowers a vegetable oil's μ about threefold. With the setpoint fixed by the stop θ_{ref}, the static balance gives $\omega_\infty = K_l\theta_{\mathrm{ref}}/(1+K_l c_s/k_s)$, so the speed varies as $1/c_s$: here $\omega_\infty = 17/(1+16\,c_s/c_{s,0})$, so a threefold drop in c_s raises the speed to $2.7\,\omega_0$ while the loop gain falls from 16 to 5.3 and the droop rises from 5.9% to 15%. This is runaway overspeed, not graceful degradation, so active compensation is not optional. A bimetallic strip closing the film gap as the oil warms helps ($c_s \propto 1/h$), but holding ω_∞ within 1% demands μ/h to about 1%—best met by a temperature-stable oil in a thermally inert chamber.
2) **Leakage and contamination.** Oil loss through the shaft seal lowers the transmitted torque and emulsion with steam condensate raises μ; both change the calibration rather than causing failure, so the governor degrades gracefully.
3) **Friction and stiction.** The film transmits only 12.7 N·m at rated speed, so bearing friction is a fixed fraction of the signal. A friction torque T_f makes the drum stick until $c_s\,\delta\omega = T_f$, so the dead band is $\delta\omega/\omega_0 = T_f/(c_s\omega_0) = T_f/12.7$ N·m. For a bronze journal of radius $r = 25$ mm, $\mu_f = 0.1$ and a normal load $N = 10$ N (the drum's weight share), $T_f = \mu_f N r = 0.025$ N·m and the dead band is 0.2% of rated speed—half the $\pm 0.5\%$ band. Hardened, oil-lubricated pivots or a jewel-and-pin arrangement are therefore required.
4) **Backlash and resolution—the least severe mode.** A 30% load step moves the drum through 1.3°, so 0.05 mm of slack in the lever pins amounts to only 0.007° at the drum; the large working travel makes backlash far less critical here than in the small-arc linkage of a flyball governor.
5) **Spring creep and aeration.** A steel torsion spring relaxes slowly under load, drifting the setpoint, and high shear entrains air that slowly reduces the transmitted torque. Periodic recalibration (or a leaf spring stressed well below its elastic limit) and a generous, baffled chamber address both.

D. Proposed Improvements

In order of value: (i) add the viscous reset dashpot of Sec. III to eliminate droop; (ii) make the film thickness h adjustable so that c_s can be re-trimmed in service; (iii) add a parallel air dashpot across the lever to raise loop damping at constant DC gain; (iv) harden and anti-backlash the lever; (v) add a temperature-compensating element against the daily swing of a mill.

REFERENCES

[1] J. C. Maxwell, "On governors," *Proceedings of the Royal Society of London*, vol. 16, pp. 270–283, 1868.
[2] S. Bennett, *A History of Control Engineering 1800–1930*. Stevenage, U.K.: Peregrinus, 1979.
[3] H. W. Dickinson and J. Rhys Jenkins, *James Watt and the Steam Engine*. Oxford, U.K.: Clarendon Press, 1927.
[4] N. S. Nise, *Control Systems Engineering*, 8th ed. Hoboken, NJ: Wiley, 2019.
[5] R. C. Dorf and R. H. Bishop, *Modern Control Systems*, 13th ed. Pearson, 2017.
[6] B. R. Munson, D. F. Young and T. H. Okiishi, *Fundamentals of Fluid Mechanics*, 5th ed. Wiley, 2006.

APPENDIX: USE OF ARTIFICIAL INTELLIGENCE

Tools used. (a) *AI assistant*: Buffy (DeepSeek V4.1 Flash, inside the Freebuff client), used as a pair-programming and drafting assistant, including the repair of LaTeX layout faults (over-long display equations, a block diagram that overflowed its column, badly stretched table cells). (b) *Typesetting*: pdfLaTeX (MiKTeX 25.12) with the `IEEEtran` conference class, `amsmath`, `booktabs` and TikZ/PGF for the schematic and the block diagram. (c) *Numerics*: MATLAB R2025b for the linear algebra, root finding, Runge–Kutta integration, frequency response and every figure; the scripts use base MATLAB/Octave functions only, so they carry no toolbox dependency. (d) Python 3 with NumPy/SciPy, used as a scratchpad for the parameter search. (e) Git and GitHub for version control. (f) Two further AI assistants, Claude and Gemini, used only as *critics* of a complete draft (see below). No computer-algebra package was needed: the linearisation was carried out by hand and verified numerically. The governing equations, the choice of every parameter and the reading of every result were the author's own; the assistant was used to draft text, to write and debug code to the author's specification, and to check algebra.

Prompts used and how each tool was used. Prompts are quoted verbatim. (1) "Derive the governing equations of a viscous drag-cup governor: a rotor on the engine shaft shearing an oil film that drags a spring-restrained sensing drum, with the drum linked to a steam throttle." (2) "Linearise these equations about $\omega_0 = 1$, $T_{L0} = 1$ and give $G_p(s)$, $G_c(s)$ and the closed-loop characteristic polynomial." (3) "Search the parameter space for a sensor stiffness, film coefficient and lever gain that give a phase margin above 60° with a steady-state speed error below 2% for a 30% load step." (4) "Write Octave-compatible MATLAB that integrates the loop with a fixed-step RK4 and produces the required transient plots." (5) "Draft an IEEE-format section on why a viscous-shear sensor is fundamentally different from a centrifugal flyweight governor." In each case the author stated the physical requirement, the tool returned a draft, and the author accepted, corrected or rejected it before it entered the paper.

Outputs accepted, modified and rejected. *Accepted after verification:* the three-block loop structure (sensor+lever, valve, engine), the RK4 integrator and the TikZ drawings. *Rejected:* the assistant's first parameter set, which placed the sensing-drum resonance below the loop crossover and gave a phase margin of only 13° with a 76% overshoot; the author rejected it and raised the drum stiffness and the lever gain until the crossover fell below both the drum and the valve corners (the design rule at the end of Sec. IV-C). *Corrected by hand:* a sign error in the disturbance transfer function and an error in the state-space realisation of the valve lag; both were found by checking the computed steady-state error against the closed-form droop $1/(1 + G_0) = 5.88\%$.

Parameters: what was fitted, and what was derived. The quantities that set the loop dynamics—K_l, c_s/k_s and J_s/k_s—were *fitted* by a numerical search that swept them until the loop met the stated targets (phase margin above 60°, steady-state error below 2% for a 30% load step under a fixed structure). The assistant ran that search. The absolute physical values were then fixed separately by dimensional reasoning from the scales ω_0, T_0 and τ_0 (Table I), and K_t was corrected to satisfy the nominal torque balance. The failure modes of Sec. VI-C were written by the author from workshop reasoning about oil, brass and lever joints. Every number quoted in the paper is regenerated by `vdg_main.m` and none of it was accepted on trust.

External review, and what it changed. A complete draft was also submitted for criticism to two further AI assistants, Claude and Gemini. Their comments led directly to several of the corrections in this revision: the K_t torque-balance error, the mis-placed T_L junction and the sensor's place in the feedback path of Fig. 2, the θ_{ss} normalisation, the film's dissipation (and the rescaling of the sensor from 637 N·m and 4 kW to 12.7 N·m and 80 W), the quantified severity of viscosity drift, the author of [3], and removal of the pressurised-leak-path mitigation, which was physically empty for a Couette film. No text was copied from those tools; each correction was checked against the model and against the scripts before it was accepted.

Figures, and what the review changed. The same review, made on the rendered PDF rather than on the source, found that the first version of Fig. 3 had been exported in the editor's dark theme (a near-black axes background with pale grey labels, unreadable in print), that its legend boxes covered the curves they labelled, that the governed colour changed from panel to panel, and that the 1.85% peak and the 1.58 s settling time were invisible on an 80 s axis. All four were corrected: every figure is now exported light-on-white with one colour and line style per case across all panels, legends are placed outside the axes, the in-figure toolbar is suppressed, and an explicit zoom panel marks the peak and the settling time. The proportional-versus-reset figure was dropped to hold the six-page limit; its content is reported numerically in Table II.